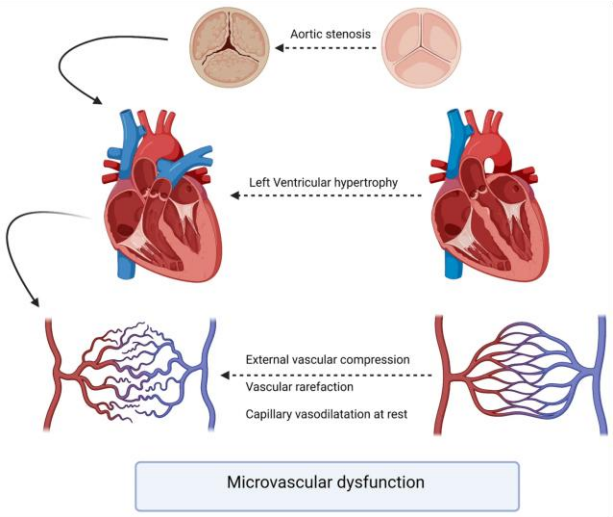
DFM-EUROPE × ELCS RESEARCH CONSORTIUM

Generative Adversarial Framework for Patient-Specific Adjustment of Fractional Flow Reserve



Scientific Context: Advanced AI models for non-invasive physiological assessment of coronary artery disease through computational fluid dynamics, deep learning, and clinical data fusion

Patient-Specific FFR Refinement Through Conditional GAN Architecture

The Challenge

Current baseline CNN models for FFR estimation provide anatomical insights from coronary imaging but **lack integration of patient-specific clinical and demographic factors** that significantly influence coronary hemodynamics.

> **Models do not account for inter-patient variability and multicentric data heterogeneity**

Research Context

- **Consortium:** DFM-EUROPE × ELCS RESEARCH scientific initiative
- **Integration:** Computational fluid dynamics, deep learning, and clinical data fusion
- **Focus:** Non-invasive physiological assessment of coronary artery disease
- **Gap:** Age, sex, comorbidities, and site variability not modeled

Why Conditional GAN?

GANs enable the generation of **physiologically plausible FFR adjustments** by learning complex, non-linear relationships between clinical variables and hemodynamic outcomes.

> **Maintains consistency with the underlying physics of coronary blood flow while personalizing predictions**

Conditional GAN Architecture: Generator and Discriminator Components

Generator (G)

Goal: Creates refined, patient-specific FFR values by adjusting the baseline CNN predictions using clinical and demographic data.

Input:

FFR-CNN baseline prediction + conditioning vector

Conditioning Variables:

- Demographic: age, sex
- Clinical comorbidities: diabetes, hypertension, obesity, prior CVD
- Data origin: multicentric site identifier

Output:

Adjusted FFR value (patient-specific refinement)

Architecture:

Uses fully connected (dense) layers with residual connections to stabilize learning and preserve input information while refining outputs.

Discriminator (D)

Goal: Evaluates whether an FFR value (real or generated) is physiologically realistic for the given patient profile.

Input:

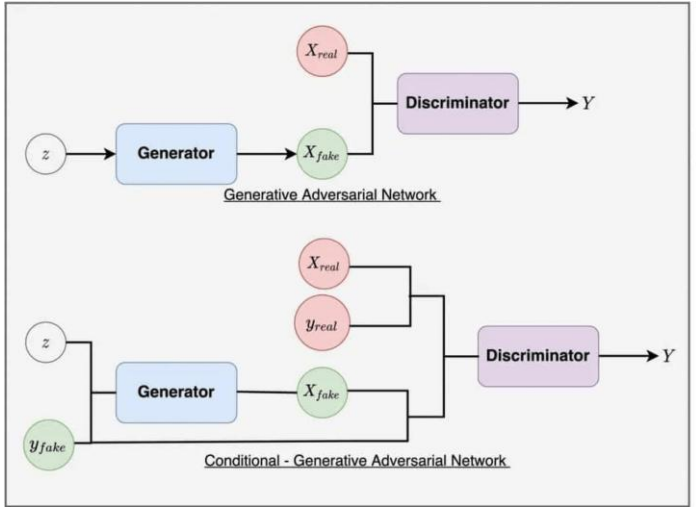
Adjusted FFR + conditioning vector OR Ground truth FFR + conditioning vector

Task:

Distinguish between generator-produced FFR and reference FFR measurements

Architecture:

Deep feedforward network with dropout for regularization to prevent overfitting



Training Strategy and Implementation Details

1. Loss Function

Multi-component objective combining:

- **Adversarial loss:** Enables realistic FFR generation through generator-discriminator competition
- **L1 reconstruction loss:** Maintains proximity to baseline FFR-CNN predictions
- **Physiological constraint loss:** Enforces domain-informed boundaries (FFR ∈ [0,1], hemodynamic plausibility)

2. Data

Multicentric coronary imaging datasets with invasive FFR as ground truth

Data Split Strategy:

- 70% training set
- 15% validation set
- 15% test set

Stratified by center to ensure balanced representation

3. Validation

Cross-center validation approach to assess model generalizability

- Test on held-out clinical centers not seen during training
- Evaluate performance across different patient populations
- Assess robustness to inter-site variability and imaging protocols

Framework Implementation

PyTorch or TensorFlow for conditional GAN architecture with iterative generator-discriminator optimization

Hyperparameter Tuning

Learning rate, batch size, loss function weights, and network depth optimized through systematic grid search

Enhanced Physiological Plausibility and Precision Cardiology

1 Patient-Specific FFR Adjustment Model

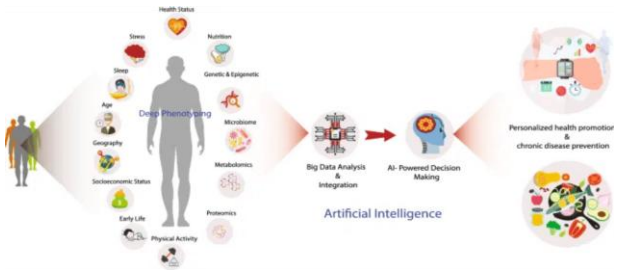
- Refined FFR predictions accounting for individual clinical profiles
- Quantified contribution of each conditioning variable (age, sex, diabetes, etc.)
- Improved alignment with invasive FFR measurements across diverse populations

2 Multicentric Variability Analysis

- Identification and quantification of inter-site differences in FFR estimation
- Data harmonization framework to reduce center-specific bias
- Enhanced model robustness for deployment across different clinical centers

3 Reproducible AI Framework

- Open-source implementation integrating data-driven learning with domain knowledge
- Modular architecture allowing incorporation of additional clinical variables
- Contribution to consortium's broader AI-based precision cardiology initiative



Clinical Relevance

→ More reliable non-invasive FFR estimation reducing unnecessary invasive catheterization

→ Personalized risk stratification for coronary artery disease management

→ Foundation for future AI-driven hemodynamic assessment tools